VERIFIED BIBLIOGRAPHIC & EMPIRICAL EXTRACT (IEA_GCCSI_CCUS_2024)
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Title: Global CCUS Status 2024 & CCUS Projects Database: Global Capacity, Sector Breakdown, and 50-Year Trajectory (1972-2024)
Authors: Global CCS Institute & International Energy Agency (IEA) (2024)
Source: Global CCS Institute Annual Report / IEA Energy Technology Perspectives CCUS Database
DOI/URL: https://www.globalccsinstitute.com/resources/global-status-of-ccs-2024/ & https://www.iea.org/data-and-statistics/data-product/ccus-projects-database
Sample: N = 628 commercial & pilot facilities tracked globally (50 operational commercial)

EXTRACTED EMPIRICAL METRICS & TABLE COORDINATES:

• Global Operational Capacity (2024): 51.0 Mt CO2/yr across 50 commercial CCS facilities (IEA tracks ~50.5 Mt/yr).
• Total Pipeline Capacity (2024): 416.0 Mt CO2/yr across 628 projects in development (+60% YoY increase; +51 Mt under construction).
• Country Breakdown (Operational): United States 22.0 Mt/yr (43.1%), Brazil 10.6-14.2 Mt/yr (Santos Basin), Canada 4.5-5.0 Mt/yr, China 4.0-5.5 Mt/yr, Norway 1.6 Mt/yr, Australia 1.6-3.4 Mt/yr, Qatar 2.1 Mt/yr, Saudi Arabia 0.8 Mt/yr, UAE 0.8 Mt/yr.
• Sector Breakdown (Operational): Natural gas processing 65.0% (33.2 Mt/yr), Hydrogen/Ammonia 12.0% (6.1 Mt/yr), Bioethanol 5.0% (2.6 Mt/yr), Power generation 3.5% (1.8 Mt/yr), Steel 1.5% (0.8 Mt/yr), Cement <0.5% (0.1 Mt/yr), DAC 0.03% (0.015 Mt/yr).
• Storage Fate: EOR represents 75-80% of historical operational capacity; Dedicated deep geological storage represents 20-25% (expanding to >70% in planned pipeline).
• Historical Milestones (1972-2024): 1972 Terrell County/Val Verde (Sharon Ridge EOR), 1982 Koch Enid, 1996 Sleipner (1.0 Mt/yr offshore saline), 2000 Weyburn-Midale, 2008 Snøhvit (0.7 Mt/yr), 2014 Boundary Dam (1.0 Mt/yr), 2017 Petra Nova (1.4 Mt/yr), 2021 Orca DAC (4 kt/yr), 2024 Northern Lights (1.5 Mt/yr) & Mammoth DAC (36 kt/yr).

VERBATIM SOURCE QUOTE / TABLE RECORD:

Global CCS Institute (2024): 'The number of facilities in operation has risen to 50 over this period, while the total number of facilities in the development pipeline has surged 60% to 628... Cumulative capture capacity reaching 416 Mtpa.' IEA (2024): 'Natural gas processing represents around 65% of operating CO2 capture capacity... commercial facilities account for ~50 Mt CO2 per year.'